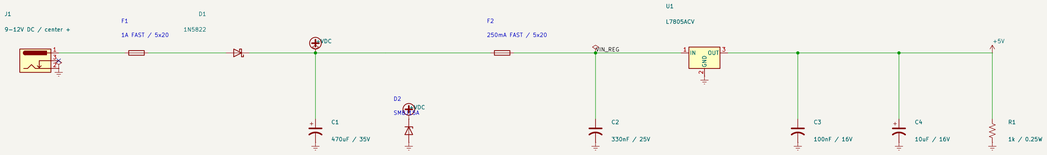
PITCH AXIS MOTOR CONTROL | NODE 2 | TEAM 103

Muhammad Ayaan Ahmad — 9–12 V input, local 5 V logic, current monitoring and latched hardware cutoff

01 POWER ENTRY, FUSES AND 5 V LINEAR REGULATION



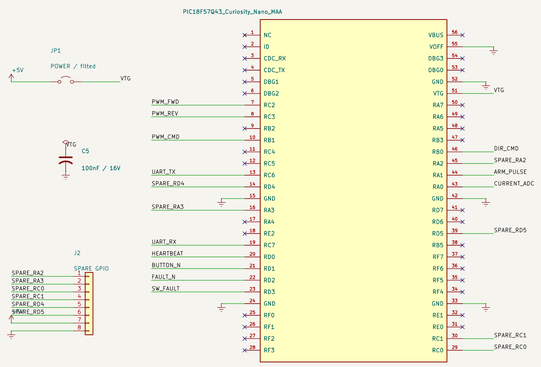
F1/F2: replaceable cartridges + two Littelfuse 100-series clips each. D1 blocks reverse input.

Use a regulator heatsink <=25 C/W. R1 guarantees >=4.8 mA minimum load. Motor power bypasses U1.

D2 and C1 absorb switching energy; validate regenerative energy at the intended motor/load.



02 CURIOSITY NANO MODULE AND EXPANSION

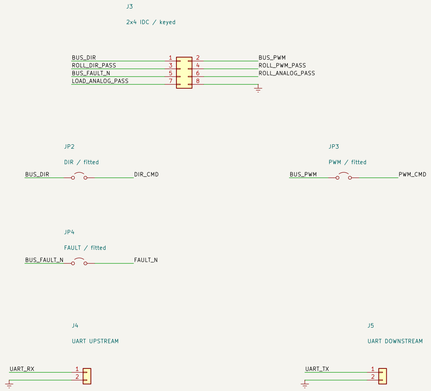


VOFF is hard grounded; VBUS is isolated. JP1 removes target power for current measurement.

USB is programming/debug only. Nano includes its own MCU bypass network; C5 bypasses VTG.

All 56 module pads shown; unused signals explicitly marked NC. Pin mapping: top view, USB upward.

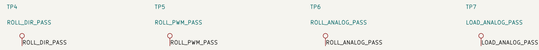
03 TEAM BUS AND UART



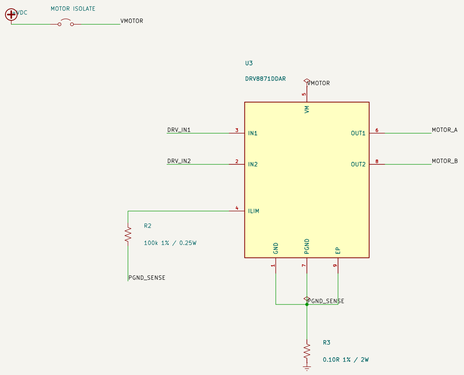
IDC 1/2 = pitch DIR/PWM, 3/4 = roll, 5 = shared active-low fault,

6 = roll analog, 7 = load analog, 8 = common. No 5 V on IDC.

Roll/load lines are monitor pads only; never driven by this board.



04 MOSFET H-BRIDGE AND LOW-SIDE CURRENT SHUNT



Bare IC; no driver daughterboard. Integrated MOSFET body diodes provide coil recirculation.

R2 sets nominal current chopping to 0.64 A. Initial continuous motor envelope: <=0.35 A.

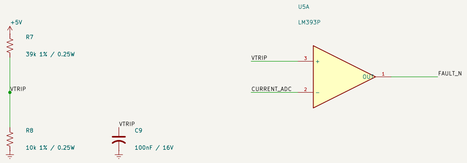
Driver GND, PGND, EP and bypass returns join ABOVE R3; use Kelvin sense routing.

Return shunt measures supply current, not recirculating winding current. Bench calibration required.

05 CURRENT AMPLIFIER, LOW-PASS FILTER AND OVERLOAD DETECTOR

Gain = 1 + 20k/1k = 21. VADC = 2.1 V/A during forward supply current.

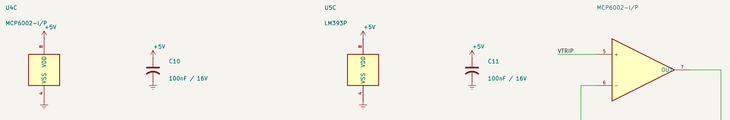
R6/C8: 1.59 kHz pole, 100 us time constant. Sample after PWM settling.



VTRIP = 5 x 10/(39+10) = 1.020 V; nominal trip about 0.486 A at DC.

Comparator pulls FAULT_N low; hardware latch holds motor OFF after current falls.

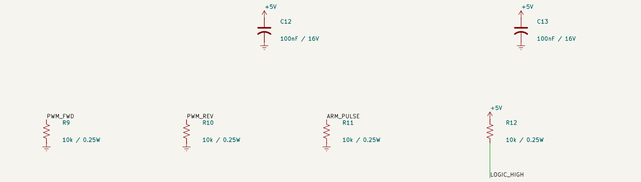
The filtered shunt is not cycle-by-cycle winding protection; DRV8871 provides that function.



Unused amplifier is a follower at VTRIP; unused comparator has defined inputs.

06 LATCHED HARDWARE SHUTDOWN

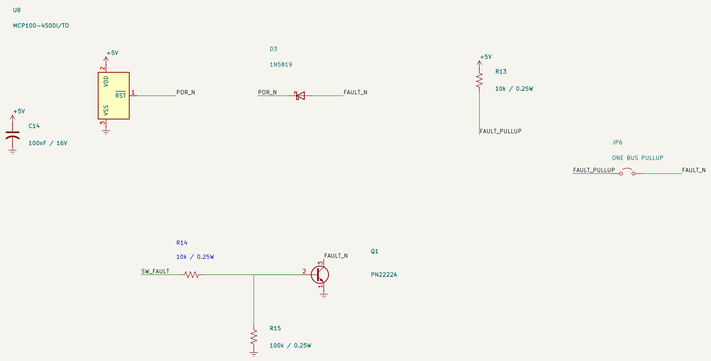




FAULT_N asynchronously clears U6. Only a new ARM rising edge can restore RUN_OK.

MCU must start with both PWM outputs LOW; arm only after explicit button action and fault checks.

07 POWER-ON RESET AND SHARED OPEN-COLLECTOR FAULT



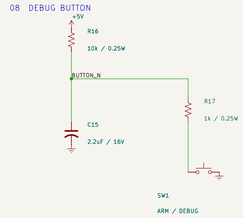
U8 + D3 hold FAULT_N low during power-up/brownout. Q1 permits MCU fault assertion.

JP6 fitted for standalone operation; team must retain exactly one shared bus pull-up.

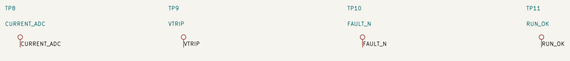
Motor: provisional 9–12 V brushed DC gearmotor, <=0.35 A continuous. Load torque, gear ratio and stall behavior must be verified.

No bench measurements claimed. Schematic based on team Product Requirements; separate UART headers preserve existing IDC assignments.

08 DEBUG BUTTON



RD1 Schmitt input: RC debounce ~22 ms release / 2 ms press.



Schematic design: motor envelope and bus integration require bench verification

RAS304 | Muhammad Ayaan Ahmad

Sheet: /

File: Pitch_Axis_MAA.kicad_sch

Title: Pitch Axis Motor Control — Team 103 Node 2

Size: A1 | Date: 2026-10-21 | **Rev: 1.0**

KiCad E.D.A. 10.0.6 | Id: 1/1